Name: ______________________

Worksheet

Rounding errors

Task A: Number lines

1) Work out the largest and smallest unrounded integers. You can use a number line if it helps.

a) 300 rounded to the nearest 10

b) 300 rounded to the nearest 100

c) 300 rounded to the nearest 50

d) 640 rounded to the nearest 10

e) 3000 rounded to the nearest 100

f) 50 rounded to the nearest 50

2) Work out the unknown number. Draw number lines to help.

a) [ ] rounded to the nearest 10

715 is the smallest integer
724 is the largest integer

b) 140 rounded to the nearest [ ]

130 is the smallest integer
149 is the largest integer

c) [ ] rounded to the nearest [ ]

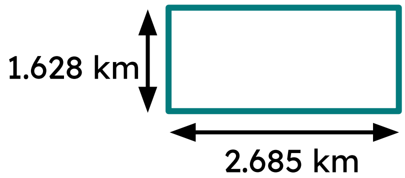
875 is the smallest integer
924 is the largest integer

3) In a school, the headteacher says there are 480 pupils rounded to the nearest 10. What is the largest number of pupils in the school?

4) An advert says 56 000 people use their product rounded to the nearest 1000. What is the least amount of people who use their product?

Task B: The impact of rounding errors

1) Here is a plot of land. Explain if and where students have lost accuracy and resulted in an incorrect answer when finding the area of the rectangle to 2 d.p.?



Aisha

$1.628 \times 2.685 = 4.38$ km^2 (2 d.p.)

Jun

$1.628 \times 2.685 = 4.37$ km^2 (2 d.p.)

Andeep

$1.63 \times 2.69 = 4.38$ km^2 (2 d.p.)

Laura

$16.28 \times 26.85 = 437.118$
$= 437.12$ km^2 (2 d.p.)

2) A circular sheet of metal has an area of 25 m^2 . It gets two square patches with lengths 2.6574 m cut from it using a laser cutter.

Which pupil has correctly worked out the unpatched area to 2 d.p.? Explain the errors from the other pupils.

Aisha	Lucas	Laura
Area of squares: $2 \times 2.6574 \times 2.6574$ $= 14.1$ (1 d.p.) Remaining area: $25 - 14.1 =$ 10.90 m^2 (2 d.p.)	Area of squares: $2 \times 2.6574 \times 2.6574$ $= 14.12$ (2 d.p.) Remaining area: $25 - 14.12 =$ 10.88 m^2 (2 d.p.)	Area of squares: $2 \times 2.6574 \times 2.6574$ $= 14.123 \ldots$ Remaining area: $25 - 14.123 \ldots =$ 10.88 m^2 (2 d.p.)